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"But they told me it was professional": Extrinsic factors in the evaluation of musical performance

Carolyn Kroger and Elizabeth Hellmuth Margulis

Abstract

This study investigated the performance preferences of listeners without formal training in music. Specifically, it asked whether the quality of the performance (as represented by the status of the performer), the order of presentation of the performances, and extrinsic information about the quality of the performance impacted preferences. In Experiment 1, participants heard pairs of performances of solo piano music and were informed that one was played by a conservatory student, and one by a world-renowned professional. After each pair, they selected the one they thought had been performed by the professional. Their responses seem to have been driven by a combination of a preference for the performance actually played by the professional and a preference for the second performance in the pair. In Experiment 2, they heard the same performance pairs, but this time were informed, correctly or incorrectly, before each performance whether it was played by a student or by a professional. After each pair, they selected the performance they preferred. This time, their responses were influenced not just by the actual performer identity and the order of presentation, but also by the priming condition. Listener preferences seem to be driven by a combination of factors intrinsic and extrinsic to the performance itself.

Keywords

aesthetics, enjoyment, familiarity, music performance, preference

Musical preferences emerge from a complex interplay among factors related both to personality and to sociocultural experiences (North & Hargreaves, 2008). Perceived quality, an evaluative response, does not always correlate with musical enjoyment, an affective response (Hargreaves, Messerschmidt, & Rubert, 1980; Thompson, 2007). An evaluative response may indicate

Music Cognition Lab, University of Arkansas, Fayetteville, AR, USA

Corresponding author:
Elizabeth Hellmuth Margulis, Music Cognition Lab, University of Arkansas, 201 Music Building, Fayetteville, AR 72701, USA.
Email: ehm@uark.edu

relative value or worth, but an affective response may relate to the level of satisfaction or reward a listener experiences (North & Hargreaves, 2001).

These issues are further complicated in the evaluation of performances of Western classical music, where people are often called upon to discriminate between multiple expressive performances of the same work. Judges at a competition may be tasked with selecting the best performance of Chopin's Second Sonata, for example, and consumers may need to choose from dozens of recordings of the piece available on Spotify, or decide whether or not to attend a particular performance of the piece. People often disagree vehemently about these choices – consider the famous case of the 1980 International Chopin Competition in Warsaw, during which Martha Argerich resigned from the jury in protest after pianist Ivo Pogorelich was eliminated in the third round. Given the possibility for disagreement, it seems clear that preference is determined not only by qualities intrinsic to a performance – elements like dynamics and expressive timing – but also by qualities extrinsic to it – factors like the personality of the listener (Rentfrow & Gosling, 2003), prior familiarity with the work (Berlyne, 1971; Heyduk, 1975; Margulis, 2013), the information provided in the program or advertisement (Margulis, 2010; Margulis, Kisida, & Greene, 2015), and perhaps even the ticket price. Even something as apparently arbitrary as the order of performance can significantly impact evaluation. Flores and Ginsburgh (1996) examined years of data from the Queen Elisabeth Competition to reveal a significant correlation between order of performance and rankings by the jury; performers who played near the end of the competition had a better chance of high rankings than performers who played near the beginning. In Juslin's (2003) model of musical expression, extrinsic factors of this type are categorized as "context-related," and are arguably the least well-understood.

Variation and convergence in performance evaluations

Expert performers with specialized musical training show substantial individual differences in preferences for musical performances. For example, when experienced performance adjudicators rated the overall quality of individual performances – some of which, unbeknownst to the judges, were identical – rating reliability for individual judges ranged from a paltry .09 to .16 (Fiske, 1978). In attempting to understand expert ratings of the quality of different performances of a single Chopin Etude, Repp (1999) could not identify a set of criteria that explained the responses; he surmised that what pianists refer to as "tone" or "touch," which is vague and difficult to specify acoustically, played a significant role; however, Repp was seeking intrinsic factors that explained differences in quality ratings. It is also possible that extrinsic factors – factors outside the internal characteristics of the performance – contributed to the judgments.

Despite these wide divergences in quality ratings, expert judgments have been shown to converge on some aspects (Bergee, 1997; Saunders & Holahan, 1997; Thompson, Diamond, & Balkwill, 1998); for example, trained listeners tend to rate more prototypical or average performances as higher in quality (Repp, 1997). Levinson (1987), however, observes that performance evaluations vary widely depending on the listener's background and aims. In other words, there is reason to suspect that people without formal musical training evaluate performances differently. Since the majority of public performance addresses a general rather than a specialized, trained audience, it is important to investigate how listeners without formal training evaluate performance.

Duerksen (1972) is one of the few studies to examine performance evaluations from listeners without special musical training. In this study, people with and without musical training listened to two expert performances of a movement from a Beethoven piano sonata. One group was told that both performances were the same; one group was told the first performance was played by a professional and the second by a student; a third group was told the opposite. Regardless of whether participants had musical training or not, they tended to prefer the second performance or the one primed as professional. This study suggests that both familiarity and priming about performer quality can impact performance preference, but leaves room to wonder about generalizability (this study used only a single performance as the stimulus) and about how these effects might interact when performances of different initial quality are used as stimuli.

Familiarity has been shown to reliably influence enjoyment, and is manipulable through repeated exposure to a stimulus. Familiarity is well-known to impact music preferences, specifically. In line with the domain-general mere exposure effect (Zajonc, 1968), preferences for musical excerpts tend to be higher after prior exposure. Once the number of exposures grows too large, however, preference begins to decline, resulting in an inverted-U shaped response (Hargreaves, 1984). As Szpunar, Schellenberg, and Pliner (2004) detail, two competing theories aim to explain this effect. On the one hand, the perceptual fluency model argues that exposure increases the ease with which people process a stimulus, but they misattribute this ease to a positive quality intrinsic to the stimulus (Bornstein & D'Agostino, 1994). On the other hand, the two-factor model argues that the initially positive effect of exposure on preference arises from an evolutionarily conditioned preference for familiarity (Berlyne, 1971). Most studies investigate the effect of familiarity on preferences for particular pieces, rather than on preferences for *performances* of individual pieces. Beyond the small-scale Duerksen study (1972), little literature has examined whether familiarity effects apply even across different performances of the same piece.

In addition to implying that familiarity effects might extend to judgments of different performances of the same piece, Duerksen (1972) also implies that the presentation of explicit information might influence performance evaluations. Because this hypothesis has been little examined in the music perception literature, it is helpful to look at the marketing literature to understand how explicit information has been shown to impact subjective evaluations in other domains.

The influence of explicit information on subjective evaluations

Outside information has been shown to influence the evaluation of items as various as cars and wine. Herr (1989) asked people to judge the price of real and fictitious cars after subtly exposing them to a collection of car names reflecting either a very high or very low price point. After being primed with a list of very expensive cars, participants judged ambiguous (fictitious) cars as more expensive, but familiar (real) cars as less expensive. Plassmann, O'Doherty, Shiv, & Rangel (2007) presented people with tastes of wine that had been labeled at different price points. People rated the subjective pleasantness of the wine's flavor higher when it was labeled as coming from a \$90 rather than a \$10 bottle. Additionally, fMRI revealed that the medial orbitofrontal cortex, a region related to experiences of pleasure, was more active when participants drank wine that had been labeled at the higher price point. Explicit information about quality, then, can transform the experience of drinking wine, enhancing or diminishing the pleasure people sustain and the ratings they assign. It might be argued that wine tasting is a more sensory and less cognitively complex activity than music listening. Can explicit information affect the enjoyment and evaluation of music performances, or will the additional cognitive and aesthetic processing involved make performances impervious to its influence?

In domains outside music, explicit information has been shown to supplement intrinsic information that is either lacking or ambiguous to increase the efficiency with which a stimulus is processed. Calling on explicit information tends to switch the principal mechanism for stimulus evaluation from bottom-up to top-down processing. In a study that analyzed whether explicit information operates on the sensory experience itself or merely the interpretation of it, participants were told either before or after drinking a beer that balsamic vinegar had been added to it (Lee, Frederick, & Ariely, 2006). People told beforehand rated their enjoyment of the beer significantly lower than people told afterward. Another group of participants were told only that a secret ingredient had been added (without revealing that the secret ingredient was balsamic vinegar). People in this group actually reported enjoying the beer significantly more. These studies reveal that extrinsic information presented before a sensory experience can impact the enjoyment people attribute to a particular stimulus. Moreover, the effect of this extrinsic information is greater for ambiguous stimuli, or stimuli with which people have less experience.

Musical stimuli can be famously ambiguous (Cross, 2005), and it is easy to identify groups of listeners who have less experience with particular repertoires; therefore, it seems plausible that music evaluations might be susceptible to the influence of extrinsic information. Even the degree to which intrinsic information contributes to musical judgments, however, is little understood. Although commonalities in performance judgments made by trained listeners have been identified (cf. Bergee, 1997; Repp, 1997), very little data exist to inform the question of how sensitive untrained listeners are to the subtle characteristics that distinguish different performances of the same piece.

The present study examines how listeners without high degrees of musical training or expertise are impacted by the presentation of extrinsic information before listening. In Experiment 1, participants heard pairs of piano performances. Each pair consisted of two performances of the same excerpt. They were asked to rate their enjoyment and the perceived quality of each, and to select which in each pair was performed by a "world-renowned professional," and which by a "conservatory student of piano." Since professionals often command high fees for their performances, and students often perform for little money or for free, this information was intended to function similarly to the price tags in Plassmann et al.'s experiment (2007), with "professional" serving as a proxy for higher quality, and "student" for lower quality. Participants in Experiment 1 judged the performances without having recourse to extrinsic information about which performance was which.

Participants in Experiment 2, however, heard the same stimuli, but were explicitly told (accurately or inaccurately) prior to each performance which one was played by the professional and which by the student. Experiment 1, in which listeners were not primed about the identity of each performance, was designed to provide a baseline against which the results of Experiment 2 could be judged. Since the setup of the two experiments was identical except for the inclusion or absence of priming about the performer's status as a student or professional, the two studies were able to isolate the impact of three factors. These three factors were the intrinsic quality of the performance (whether it was actually played by a student or a professional), as well as two extrinsic factors: familiarity (whether they preferred the second performance in each pair); and verbal information about the performance's quality (priming about the performer's status).

If ordinary listeners are able to distinguish between professional and student performances on the basis of qualities intrinsic to the performance, they should prefer the professional performances regardless of whether they occur first or second within the pair, and regardless of how they are primed. If extrinsic factors play a role, however, participants should prefer the second performance in each pair, reflecting the increased familiarity brought by the prior exposure to the piece, and they should prefer the performance primed as professional, regardless of whether it was actually played by a student or a professional.

Experiment 1

Method

Participants. A total of 40 undergraduate students (21 male and 19 female), aged 18–22 years ($M = 19.2$, $SD = 1.2$) participated in exchange for credit in a general psychology course at the University of Arkansas. None of the participants were music majors. Eight participants had prior training in piano with experience ranging from 2 to 13 years ($M = 4.1$, $SD = 3.7$), but none played the piano currently. The data was analyzed with and without the eight participants who reported some form of prior piano training included; since the results were not significantly different, all participants have been included in the analysis reported here.

Materials. Stimuli were 90–120 second segments of 8 piano pieces from the common practice period (see Appendix 1). For each piece, one performance by a conservatory student and one by a world-renowned professional were selected as stimuli. Student performances were drawn from user-uploaded recordings on the internet as well as email submissions from graduate and post-graduate students of piano. Professional performances were drawn from commercially available recordings. The excerpts were specifically selected in order to minimize differences in recording quality. The excerpts were trimmed so that they started and ended at the same moment in each piece, and were edited to make volume levels and static noise consistent across recordings.

As shown in Table 1, the stimuli were presented in 8 different pairs. Within each pair, they heard the same piece twice. For two pairs, the professional recording was played first and the student second (pieces 2 and 6); for two pairs the student recording was played first and the professional second (pieces 1 and 5); for two pairs the professional recording was played twice (pieces 4 and 8); and for two pairs, the student recording was played twice (pieces 3 and 7). The order in which the pairs were presented was randomized for each participant.

Participants were asked to rate their enjoyment and the performer's skill level on 7-point Likert-like scales. They were also asked a number of questions adapted from a previous music preference study (Repp, 1997). They were familiarized with definitions of tempo ("The speed or pace of the music, how fast or how slow"), dynamics ("the relative volume of the notes, how loud or how soft"), and expression ("the ability for the music to communicate or convey emotion or meaning"). Then, they were asked to rate various aspects of the performance on 5-point Likert-like scales where the middle value was labeled as "just right," and steps to the left and right of the middle value indicated progressively less ideal performances. For example, the options on the tempo scale were "much too slow," "too slow," "just right," "too fast," and "much too fast." Thus, the middle value represented the highest rating, and the outer values represented the lowest. The dynamics scale ranged from "much too small" to "much too large;" expressivity from "inexpressive" to "exaggerated;" and conventionality from "much too conventional" to "much too unconventional."

Procedure. Upon arrival, participants signed a consent form and were ushered into a Whisper-Room 4' by 4' Enhanced, Double Wall Isolation Booth, where they sat at a computer terminal outfitted with Sennheiser HD 600 headphones. They first completed a demographic questionnaire. Prior to the presentation of the main experimental trials, they were told that they would

Table 1. Order of presentation of performances.

Piece Number	Condition Name	Performance 1	Performance 2
1	SP	Student	Professional
2	PS	Professional	Student
3	SS	Student	Student
4	PP	Professional	Professional
5	SP	Student	Professional
6	PS	Professional	Student
7	SS	Student	Student
8	PP	Professional	Professional

be hearing 8 pairs of piano performances. Within each pair, they were told they would hear two performances of the same piece, where one would be performed by a "conservatory student of piano" and the other by a "world-renowned professional pianist" (although in reality, some of the time they were hearing the same performance twice, as described above). Participants then proceeded to hear the 8 pairs of excerpts in random order. After each individual excerpt (16 total), they provided evaluations on the Likert-like scales described above. Following each pair, they were asked to choose which of the two excerpts was performed by the "world-renowned professional."

Results

The data were analyzed with a repeated measures logistic regression using Generalized Estimating Equations (GEE) to test for effects of performer type (student or professional) on the likelihood that participants would choose the either the first or second performance as the professional in each of the four conditions.

Performer type had a significant effect on performance choice, Wald χ^2 ($df = 3$, $N = 40$) = 15.8, $p < .001$; that is, people were more likely to say that a performance was played by a "world-renowned professional" when it actually was. Familiarity also played a role, with participants labeling the second performance as the "world-renowned professional" 62.8% of the time – 12.8% higher than expected had they been choosing at random. Overall, participants' responses seem to have been modulated by a combination of this familiarity effect and the actual performer type. As shown in Figure 1, about 70% of participants chose the second performance when the performer type of the second performance was indeed professional; this number decreased to 60% when neither performer type was professional, and to 50% when the first performer type of the first performance was professional.

When measures of enjoyment were included in the regression model, they significantly predicted which performance was chosen as professional. Enjoyment ratings of the second performance had a coefficient of $b = -1.7$ ($OR = 0.17$), Wald $\chi^2 = 60.8$, $p < .001$. As enjoyment ratings for the second performance increased by 1 point, the odds of choosing the first performance as the professional decreased by .82 or 82%. Quality ("skill level") ratings also predicted the choice of professional, with a quality rating for a coefficient of $b = -2.2$ ($OR = 0.1$), Wald $\chi^2 = 40.5$, $p < .001$. As quality ratings for the second performance increased by 1, the odds of choosing the first performance as the professional decreased by .89 or 89%. Additionally, enjoyment and quality ratings correlated strongly; when people rated a performance as highly enjoyable, they also tended to rate the performer as highly skilled, $r(320) = .65$, $p < .001$. Enjoyment

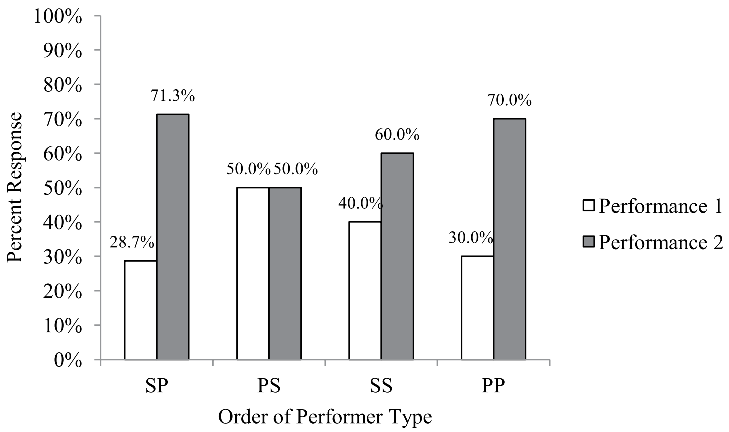

Figure 1. Percentage of participants who chose the first and second performance as professional in each condition in Experiment 1.

and quality ratings themselves did not differ significantly as a function of performer type or familiarity.

Expressivity ratings were made on a 5-point scale where 3 indicated "just right," and distance from this central value represented progressively poorer ratings. For analysis, these ratings were converted to a 3-point scale, where 3 ("just right") represented the highest value, 2 and 4 the next highest (e.g. "a little too fast" or "a little too slow"), and 1 and 5 the least (e.g. "much too fast" or "much too slow"). These transformed values represented "distance from just right," where the particular direction of the difference (whether it was too fast or too slow, for example) did not matter. When tempo was included in the regression model, ratings for the second performance had a coefficient of $b = -1.4$ ($OR = 0.24$), with Wald $\chi^2 = 26.2$, $p < .001$. As tempo ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .75 or 75%. When dynamic ratings were included in the model, ratings for the second performance had a coefficient of $b = -2.0$ ($OR = 0.12$), with Wald $\chi^2 = 47.6$, $p < .001$. As dynamic ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .87 or 87%. Expressivity was also included in the model and ratings for this parameter for the second performance had a coefficient of $b = -1.5$ ($OR = 0.2$), with Wald $\chi^2 = 21.4$, $p < .001$. As ratings of expressivity for the second performance increased by 1, the odds of choosing the first performance decreased by .79 or 79%. Finally, individuality ratings for the second performance had a coefficient of $b = -1.5$ ($OR = 0.2$), with Wald $\chi^2 = 32.0$, $p < .001$. As individuality ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .77 or 77%. The results for these additional parameters indicate that not only do quality and enjoyment increase with exposure, but specific expressive attributes are also rated significantly higher upon the second listen (see Figure 2).

Discussion

Participants labeled the second performance as professional more frequently than the first performance, but this effect was modulated by the actual identity of the performer. Listeners' evaluations seem to have been driven by a combination of a preference for performances of pieces that were already familiar from a prior exposure, and a preference for performances that were

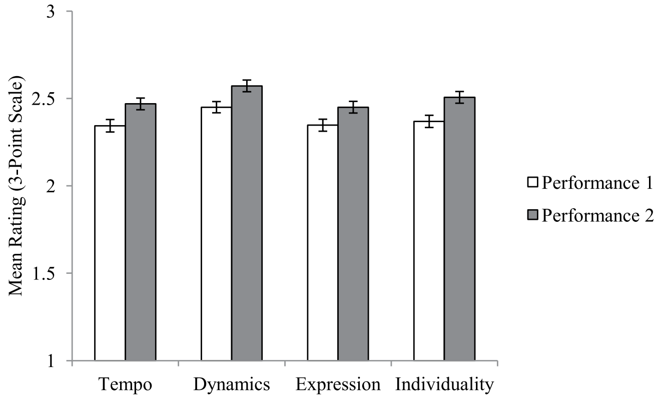

Figure 2. Mean ratings for expressive parameters in Experiment 1.

actually played by professionals rather than students. This reflects a combination of extrinsic and intrinsic factors in performance evaluation, supporting Juslin's (2003) model of musical expression.

Participants rated the performance they chose as professional as both more enjoyable and of higher quality than the other one, even when both performances were actually the same (as in the SS and PP conditions). Reporting different levels of enjoyment and quality across two exposures to the same piece suggests that participants were strongly influenced by extrinsic factors. It also suggests that participants associated the label "professional" with performances that were higher quality and more enjoyable, just as participants in Plassmann et al. (2007) associated higher price points with better and more enjoyable wine.

The differences relating to performer type and familiarity were evident in the selection of which performance was played by the professional, but not in the direct ratings of enjoyment and quality. This might suggest that the differences are subtle: detectable only by a forced choice rather than direct rating paradigm. But it might also suggest that people without formal musical training are more accustomed to the task of choosing between two performances, which more closely approximates real world consumer choices, than explicitly rating the performances themselves.

Although familiarity has long been known to impact music preferences (as well summarized in Szpunar et al., 2004), this study shows that prior exposure to a piece can be enough to make listeners think a new performance of it is actually better. The results from this study also suggest that familiarity has an impact on the perceived quality of a performance. People may rate specific technical components, such as tempo, dynamics, expressivity, and individuality, as more successfully executed on second exposure to a particular piece. The mere exposure effect can influence not only ratings of a piece, but also ratings of a performance, such that people presented with two performances of the same piece will tend to rate the second as superior.

Experiment 2

Method

Participants. A total of 40 undergraduate students (13 male and 27 female), aged 18–32 years (M = 20.0, SD = 3.1) participated in exchange for credit in a general psychology course at the

Table 2. Presentation order and priming conditions.

Piece Number	Prime 1	Prime 2	Performance 1	Performance 2
1	Professional	Student	Student	Professional
2	Student	Professional	Professional	Student
3	Professional	Student	Student	Student
4	Professional	Student	Professional	Professional
5	Student	Professional	Student	Professional
6	Professional	Student	Professional	Student
7	Student	Professional	Student	Student
8	Student	Professional	Professional	Professional

University of Arkansas. Nine participants reported prior training in piano, with experience ranging from 2 to 11 years ($M = 5.0$, $SD = 3.0$), and one participant was a music major. Participants from Experiment 1 were not eligible to participate in Experiment 2. The data was reanalyzed without the nine participants who had prior piano training as well as the one music major, but since the results did not change, all participants were included in the analysis discussed here.

Materials. The materials in Experiment 2 were the same as those in Experiment 1, except for the addition of a verbal priming element. Instead of being told that they would hear a pair of performances where one was a student and one was a professional, instructions explicitly indicated before each performance within the pair whether it was the performance by a "world-renowned professional" or a "conservatory student." However, as shown in Table 2, the primed identity was only congruent with the actual identity of the performer 50% of the time. There were two excerpt pairs in each of the four presentation orders (Student–Professional; Professional–Student; Student–Student; and Professional–Professional); for one of the pairs, participants were told the first performance was played by a professional and the second by a student, and for the other, they were told the first was played by a student and the second by a professional. As an example, consider the two excerpt pairs with the presentation order Professional–Student – Excerpt 1 and Excerpt 6, as listed in Table 2. For Excerpt 1, participants were told (inaccurately) that the first performance was played by a student and the second by a professional. For Excerpt 6, however, they were told (accurately) that the first performance was played by a professional and the second by a student.

Procedure. The procedure in Experiment 2 was the same as that in Experiment 1 except for a change in instructions that specified whether each piece was performed by a student or a professional (information that was sometimes accurate and sometimes false). During the listening portion of the experiment, participants were presented with the same 8 pairs from Experiment 1 in randomized order, and were either falsely or correctly informed about which performance they were hearing (see Table 2). They completed the same questionnaires after each performance as in Experiment 1, except that they were not asked which performance was the professional, since this information had already been provided. Instead, they were asked: "Which performance did you prefer?"

Results

As in Experiment 1, a GEE was used to perform a logistic regression, this time testing for the effects of priming condition and performer type on the likelihood that either the first or second

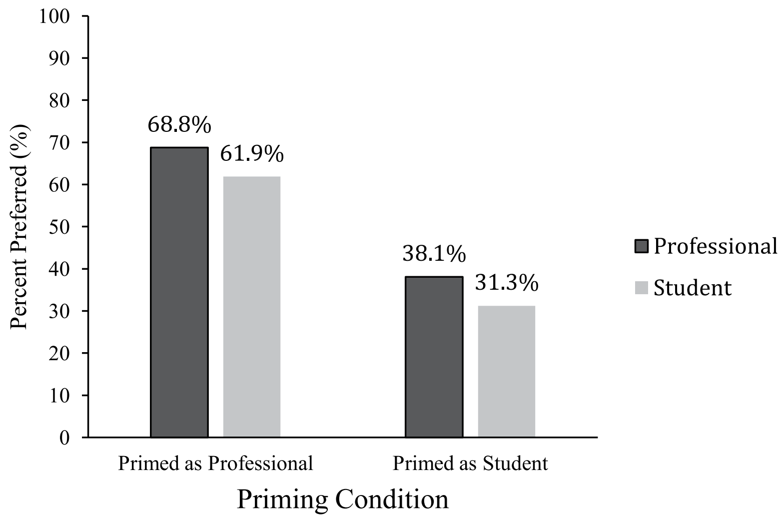

Figure 3. Percentage of participants who preferred professional and student performances when they were primed as professional or student.

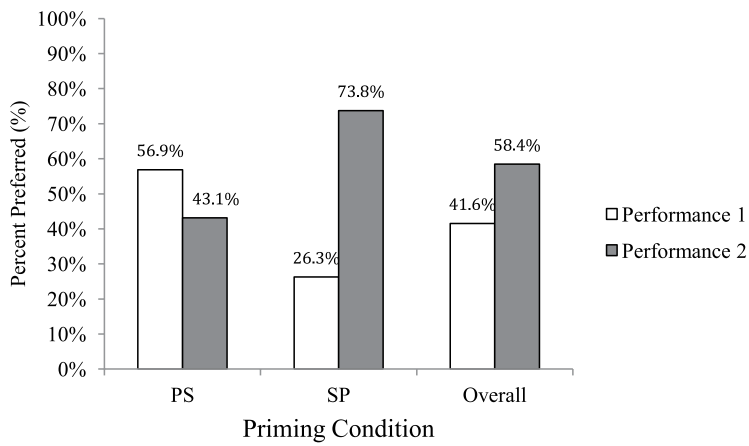

Figure 4. Percentage of participants who preferred Performance 1 and Performance 2 by priming condition.

performance was preferred. There was a significant effect of priming condition, Wald χ^2 ($df = 3$, $N = 40$) = 23.7, $p < .001$, and a smaller significant effect of performer type, Wald χ^2 ($df = 3$, $N = 40$) = 8.3, $p = .04$, but no significant interaction between the two, Wald χ^2 ($df = 3$, $N = 40$) = 3.6, $p = .305$. As shown in Figure 3, when one performance was primed as a professional and the other was primed as a student, participants most often preferred the performance primed as a professional.

Overall, participants preferred performances labeled as the "world-renowned professional" 65.3% of the time, 15.3% higher than predicted by chance. As shown in Figure 4, when the first excerpt was primed as professional, participants preferred it 56.9% of the time, but when the second excerpt was primed as professional, participants preferred it 73.8% of the time,

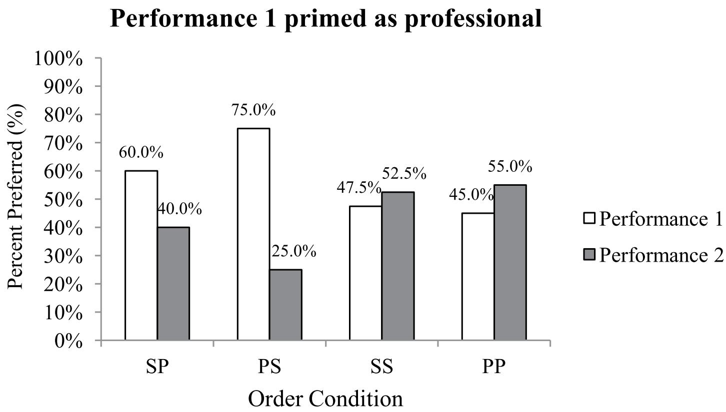

Figure 5. Percentage of participants who preferred Performance 1 or Performance 2 for the priming condition in which professional was primed first and student was primed second.

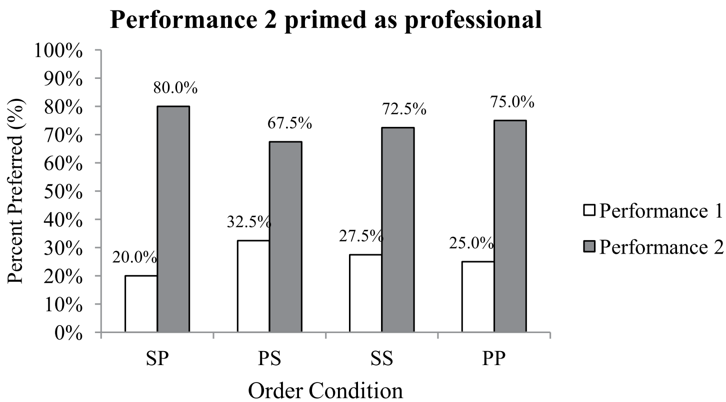

Figure 6. Percentage of participants who preferred Performance 1 or Performance 2 for the priming condition in which student was primed first and professional was primed second.

23.8% higher than predicted by chance. Preferences were driven not only by priming condition, but also by presentation order (see Figure 5 and 6).

Participants preferred the first performance (70%) when the professional prime was congruent with the performer type, but less so (60%) when it contradicted the performer type. When a pair of performances shared the same performer type (SS or PP), and the first was primed as professional, participants chose the first and second performance with approximately equal frequency (see Figure 5). Since performer type is controlled for in these conditions (both performances are the same), this distribution reflects a tension between priming (which favors the first performance), and the effect of familiarity (which favors the second).

When the second performance was primed as the professional, results looked quite different (see Figure 6). In these cases, the effects of familiarity and priming were working together rather than against each other, resulting in high preferences for the second performance in all order conditions. The highest percentage of participants (80%) chose the second performance when it was not only primed as professional, but the performer type was also professional. This number dropped to 67.5% when the second performance was primed as professional, but only the first performance had the professional performer type. About three-quarters of participants chose the second performance as professional when there was no change in performer type between the two (SS and PP conditions).

When measures of enjoyment and quality were included in the model, they significantly predicted which performance participants chose as the professional. Ratings of quality ("skill level") for performance 2 had a regression coefficient of $b = -1.087$ ($OR = 0.33$), Wald $\chi^2 = 32.0$, $p < .001$. As quality ratings for the second performance increased by 1 point, the odds of preferring the first performance over the second one decreased by 66%. Enjoyment ratings exhibited a similar trend where the coefficient for performance 2 was $b = -1.6$ ($OR = 0.19$), Wald $\chi^2 = 25.1$, $p < .001$. Thus, enjoyment ratings predicted performance choice even more strongly than did quality ratings. As enjoyment ratings for the second performance increased by 1, the odds of preferring the first over the second performance decreased by 81%. Enjoyment and quality were strongly correlated where the Pearson coefficient for performance 1 was $r = 0.6$, $p < .001$ (2-tailed), and for performance 2 was $r = .5$, $p < .001$ (2-tailed). Neither enjoyment nor quality ratings varied significantly depending on presentation order, performer type, or priming condition.

When tempo was included in the regression model, ratings for the second performance had a coefficient of $b = -2.1$ ($OR = 0.1$), with Wald $\chi^2 = 31.4$, $p < .001$. As tempo ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .89 or 89%. When dynamic ratings were included in the model, ratings for the second performance had a coefficient of $b = -2.2$ ($OR = 0.1$), with Wald $\chi^2 = 57.2$, $p < .001$. As dynamic ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .89 or 89%. Expressivity was also included in the model and ratings for this parameter for the second performance had a coefficient of $b = -1.4$ ($OR = 0.2$), with Wald $\chi^2 = 20.9$, $p < .001$. As ratings of expressivity for the second performance increased by 1, the odds of choosing the first performance decreased by .77 or 77%. Finally, individuality ratings for the second performance had a coefficient of $b = -1.5$ ($OR = 0.2$), with Wald $\chi^2 = 20.5$, $p < .001$. As individuality ratings for the second performance increased by 1, the odds of choosing the first performance decreased by .80 or 80%.

Discussion

Results for Experiment 2 demonstrate an interplay between intrinsic and extrinsic factors in the evaluation of musical performance, once more supporting Juslin's (2003) model of musical expression. When people were told a performance was played by a "world-renowned professional," they preferred it over a performance they were told was played by a "conservatory student." However, this impact was mitigated by the influence of performer type and familiarity. When the first performance was primed as professional, people preferred it in the SP and PS conditions, but selected it only half the time in the SS and PP conditions; contrastingly, when the second performance was primed as professional, people preferred it in all four conditions. Additionally, the margin by which people preferred the performance primed as professional was larger when that performance actually was performed by the professional (75% versus 60% when it was the first performance, and 80% versus 67.5% when it was the second).

According to classic models of expressive performance, intrinsic qualities of a performance – acoustic phenomena such as microtiming and dynamic fluctuations – determine listener response (see Gabrielsson, 1999 for a summary). The fact that performer type significantly influenced participant ratings confirms this model to a certain extent; participants were presumably able to pick up on acoustic features that distinguished professional from student performances, even in the absence of any such labels. Yet these classic models cannot be complete accounts, because extrinsic factors shaped responses as well – an idea put forward by Juslin (2003). Specifically, familiarity – already known to influence preference for individual recordings – was shown here to influence experiences of a second performance of an individual piece, such that listeners interpreted the second performance as higher quality and more enjoyable. Additionally, the presentation of verbal information about the performance, namely labeling it as the result of professional or student work, shifted responses such that listeners tended to enjoy the one labeled as professional more.

General discussion

Participants' evaluation of musical performances was influenced by the order in which they heard them, the actual identity of the performer (whether professional or student), and what they were told about the performer. Despite that these participants for the most part lacked specialized training in music, they still on the whole showed a preference for performances actually played by professionals over performances actually played by students (Experiment 1). Given that the student performances were all of conservatory quality, this is an impressive feat of discrimination. Juslin (2003) introduced the notion that in addition to context-related factors (such as the setting in which the performance is heard), listener-related factors (such as the expertise of the listener) influence the experience of musical expression. In the study reported here, prior expertise is shown not to be necessary in order to discriminate professional from student performances at greater than chance rates. These preferences were strongly modulated, however, by the information they were provided (Experiment 2): if participants were told a performance was by a professional, they tended to prefer it, even if that performance was actually by a student. Both of these effects were influenced by a third factor: the order in which the performances were presented. People tended to prefer the second performance in each pair.

These results demonstrate that aesthetic evaluations are formed by a combination of qualities intrinsic to the stimulus, and qualities extrinsic to it, including familiarity with the piece and explicit information about the performer. Margulis (2010) demonstrated that program notes can have a negative impact on musical enjoyment, but Margulis et al. (2015) suggested that they can also have a positive impact on listeners for whom the concert-going experience is unfamiliar. In both of these studies, the program notes conveyed information about the content of the work, rather than signals about the quality of the performer. This study shows that explicit information about performance quality can prime listeners such that they have a more enjoyable experience of the music, regardless of variations in intrinsic quality. Taken together, these studies suggest that the information audience members receive about a performance can significantly alter their experience. Given the outreach efforts currently being undertaken by major orchestras and concert presenters (John S. and James L. Knight Foundation, 2002), this dimension of the listening experience seems particularly worthy of further exploration. Additionally, it has often been argued that musical experiences include significant nonconceptual content (DeBellis, 1995) or that parts of them are ineffable, or resistant to verbal capture (Raffman, 1993). Therefore, when verbal information is shown to shape musical experiences, the implications for broader questions about the relationship between music and language (Patel, 2008) and the way information within one channel (e.g. language) can reconstitute information in another (e.g. music) are particularly worth pursuing.

In these experiments, effects were noticeable when participants were forced to choose which of two performances was played by a professional, or which of two performances they preferred, but not when they individually rated the quality or enjoyment of each performance. Aesthetic experiences can be notoriously difficult to articulate, and it could be that while participants had a difficult time making explicit ratings based on specific attributes, they were able to rely on intuition to select one performance over the other. This distinction could suggest that future studies in the psychology of aesthetics might aim to use forced choice rather than explicit rating type tasks. Given the resurgence of interest in empirical aesthetics, demonstrated by the recent founding of the Max Planck Institute for Empirical Aesthetics and the revitalization of two journals – *Empirical Studies of the Arts* and the *Psychology of Aesthetics, Creativity, and the Arts* – this kind of methodological support for measures that avoid reliance on verbalization can be of broad relevance, helping reveal implicit processes that shape listening for people without formal musical training who are unable to put their experiences into words.

Future research should investigate whether listeners with extensive formal training are subject to the same effects. Perhaps their evaluations would depend more strongly on the actual performer type, and less on the extrinsic factors of order and priming. It would also be helpful for future research to add some indirect measures of enjoyment, such as neuroimaging or electrophysiological approaches.

This study has implications for many areas related to aesthetic evaluation and preference including performance judgment, concert program notes, marketing and advertising. If being informed that a performance is played by a student or professional can modulate enjoyment of it, then perhaps program notes or advertisements that emphasize a performer's renown can actually impact the pleasure people derive from a concert. Performers seem to sense this intuitively; there is certainly no shortage of pianists who refer to themselves in publicity materials as "world-renowned." The familiarity effects may be applicable to the judgment of music competitions because performances, sometimes of the same piece, are heard by judges in succession (Biswas, Grewal, & Roggeveen, 2010; Biswas, Labrecque, Lehmann, & Markos, 2014). These competitions often determine the future of a performer, and have profound impacts on acceptance into various music programs or schools as well as success in the classical music industry (Bruine de Bruin, 2005). The priming effects displayed in Experiment 2, paired with the results of previous research (Margulis, 2010; Margulis et al., 2015), may inform how program notes and marketing detract from or contribute to the experience of a live performance. In general, they point to the possibility of a new research area in the psychology of aesthetic experience.

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Appendix 1. Stimuli.

Piece Number	Piece Name	Composer	Professional Performer	Student Performer
1	Piano Sonata in E, Op. 109, II	L. V. Beethoven	Alfred Brendel	Student 1
2	Piano Sonata in B Flat, K. 570, II	W. A. Mozart	Mitsuko Uchida	Student 2
3	Intermezzo in E, Op. 116, No. 4	J. Brahms	Julius Katchen	Student 3
4	Sonata No. 2 in B Flat Minor, Op. 35, I	F. Chopin	Leif Ove Andsnes	Student 4
5	Prelude in B Minor No. 10, Op. 32	S. Rachmaninoff	Vladimir Ashkenazy	Student 5
6	Paganini Variations	F. Say	Fazil Say	Student 6
7	From a Log Cabin Op. 62, No. 9	E. Macdowell	James Barbagallo	Student 7
8	Sonata in G Minor, K. 426	D. Scarlatti	Nikolai Demidenko	Student 8